



FIELDPROOF™

Sensor-Verified Human Task Execution

Nestor Villalobos

Instructor: Dr. Sina Nazifi | ITAI 1371
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 **IMPROVE
SAFETY**

 **ENSURE
COMPLIANCE**

 **REDUCE
DOWNTIME**

 **DRIVE
EFFICIENCY**

 **GO
DIGITAL**



PROVING WORK. PROTECTING PEOPLE. POWERING PERFORMANCE.

FIELDPROOF™

VERIFIED WORK. PROVEN COMPLIANCE. SAFER OPERATIONS.

FIELDPROOF™ is a revolutionary **proof-of-compliance** system that transforms how oil and gas operations verify, document, and authorize critical field work.

THE PROBLEM



Manual documentation is inconsistent and unreliable



Procedural non-compliance leads to incidents



Increased liability, downtime, and regulatory risk



Paperwork delays and high workload strain oversight

THE SOLUTION



Wearable sensors and IoT devices capture real-world activity



AI verifies work against safety and operational protocols



Automated logging creates audit-ready digital proof



Instant verification and approvals ensure work progresses safely



REAL WORK.
VERIFIED.
AUTOMATED.
TRUSTED.

FIELDPROOF exists because real work needs real proof — not paper, not promises.

Oil & gas, maritime, and industrial operations still rely on manual compliance — paper checklists, supervisor sign-offs, and self-reported procedures that are slow, inconsistent, and easy to falsify.

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AUTOMATED VERIFICATION

Wearable sensors and IoT devices capture real-world activity and validate that procedures are followed correctly.



REAL-TIME COMPLIANCE

Instant feedback and alerts ensure issues are addressed immediately, before they become incidents.



AUDIT-READY EVIDENCE

All verified data is time-stamped, immutable, and stored securely for regulatory and internal audits.



ELIMINATE PAPERWORK

Goodbye to manual checklists, signatures, and spreadsheets. Hello to digital proof that stands up to scrutiny.



IMPROVE SAFETY & EFFICIENCY

Reduce incidents, increase first-time-right work, and empower your workforce.

SENSOR-VERIFIED PROOF AT EVERY STEP



WEARABLE SENSORS

- Body Movement
- Posture & Gesture
- Heart Rate (optional)
- Location (GPS)



EQUIPMENT IOT DEVICES

- Tool Usage
- Equipment Status
- Pressure / Temperature
- Vibration / Runtime



ENVIRONMENTAL DATA

- Gas Detection
- Weather Conditions
- Light / Noise / Air Quality



TASK INTERACTION

- Valve Turn
- Gauge Check
- Inspection Photo
- Component Replacement



REAL WORK. VERIFIED. AUTOMATED. TRUSTED.

FIELDPROOF turns field activity into undeniable **proof-of-compliance**.

Proof of Compliance

System that transforms how oil and gas operations verify, document, and authorize critical field work.

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HOW FIELDPROOF WORKS

1 TASK ASSIGNMENT



Work orders and procedures are assigned to the worker's device.

2 TASK EXECUTION



Worker performs tasks using proper procedures with wearable & IoT sensors active.

3 DATA CAPTURE



Movement, equipment, environmental, and interaction data are captured in real-time.

4 AI VERIFICATION



Machine learning models analyze data to verify correct steps, sequence, and conditions.

5 AUTOMATED LOGGING



Verified results are automatically logged with time-stamped, immutable records.

6 APPROVAL & PROGRESSION



Compliance confirmed. Approvals triggered. Work progression authorized.

KEY BENEFITS



REDUCE INCIDENTS

Ensure procedures are followed every time.



IMPROVE WORKFORCE PRODUCTIVITY

Less rework, fewer delays, better execution.



STRENGTHEN REGULATORY COMPLIANCE

Always audit-ready with verifiable evidence.



LOWER OPERATIONAL COSTS

Eliminate paperwork, minimize downtime.



ENABLE DIGITAL TRANSFORMATION

Real data. Real insights. Smarter operations.

BUILT FOR THE OIL & GAS INDUSTRY



WELL OPERATIONS



INSPECTIONS



MAINTENANCE



TURNAROUNDS



CONSTRUCTION

TRANSFORMING OILFIELD OPERATIONS



From manual checklists to digital proof



From supervisor signoffs to AI-verified compliance



From reactive to proactive safety



From siloed data to real-time insights



From compliance burden to competitive advantage



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Why this problem?

Oil & gas, maritime, and industrial operations still rely on manual compliance paper checklists, supervisor sign-offs, and self-reported procedures that are slow, inconsistent, and easy to falsify.

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1 Motivation

Why this problem?

Oil & gas, maritime, and industrial operations still rely on manual compliance paper checklists, supervisor sign-offs, and self-reported procedures that are slow, inconsistent, and easy to falsify.

Human Error & Falsification

Workers skip steps or mark tasks complete without doing them. Paper offers no proof.

Procedural Incidents

30–40% of offshore safety incidents trace back to missed or out-of-sequence task steps.

Slow Audits

Post-task documentation takes hours. Audits are reactive, not preventive.

Wasted Resources

Supervisors spend up to 25% of their shift verifying work manually.

\$500M+

Annual cost of offshore non-compliance incidents

40%

Tasks with falsified or missing completion records

1 in 3

Industrial accidents linked to procedural failures

A better, automated verification system is needed to improve safety, reduce incidents, and increase operational efficiency.

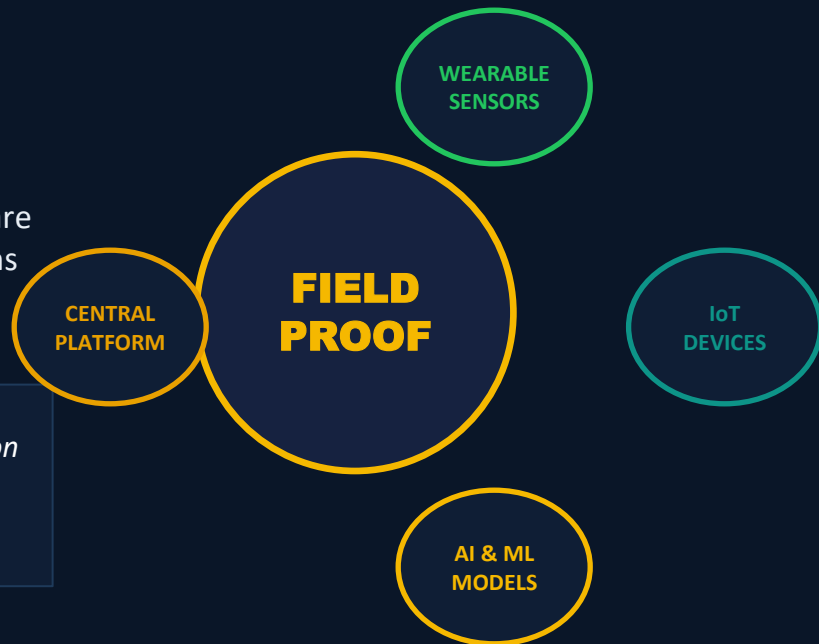
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2 Introduction

What is FIELDPROOF?

FIELDPROOF is a proof-of-compliance system that uses wearable sensors, IoT devices, and machine learning to verify that field tasks are completed correctly, in the right sequence, and under safe conditions — without any manual documentation.

It replaces manual documentation with real-time, sensor-based validation — ensuring procedures are followed in the correct sequence and conditions every time.



Problems Solved

- ✓ Falsifiable paper records replaced with sensor-verified proof
- ✓ Sequential task verification in real time
- ✓ Auto-logging to ERP, HSE, and audit systems
- ✓ Worker accountability without invasive surveillance

Worker accountability without invasive surveillance

Key Use Cases

- Offshore rig lockout/tagout verification
- Maritime cargo lashing compliance
- Refinery confined-space entry checks
- Hospital procedure compliance (CLINICPROOF)

3 Background

Traditional compliance systems depend on manual reporting and post-task audits.

TRADITIONAL APPROACH

- ✗ Manual checklists
- ✗ Supervisor sign-offs
- ✗ Paperwork & spreadsheets
- ✗ Delayed post-task audits
- ✗ High risk of errors & falsification

vs

STATE-OF-THE-ART SOLUTIONS

- ✓ IoT sensors for real-time data collection
- ✓ Digital tracking & workflow tools
- ✓ Real-time visibility dashboards
- ✓ Better data management & analytics

But they track data, not actual execution.

FIELDPROOF goes beyond tracking — it provides verifiable **proof-of-work** through human activity and sensor data.

4

Methods

A simulated FIELDPROOF dataset was created using biometric and activity-based features from field operations.



Wearable
Sensors



Activity &
Movement



Environmental
Conditions



Equipment
Interaction



Task
Metadata

OUR APPROACH



Data Collection
(Simulated)



Data Preparation
& Feature Eng.



Machine Learning
Models



Compliance
Prediction



Models Used: Logistic Regression, Random Forest, Decision Tree, K-Means, PCA | **Evaluation:** Accuracy, Precision, Recall, F1-Score, Fairness Audit (FPR/FNR)

5 Results

The models demonstrated strong ability to predict compliance outcomes across all three classes.

MODEL PERFORMANCE

✓ Accuracy **93.3%**

✓ Precision **90.6%**

✓ Recall **89.4%**

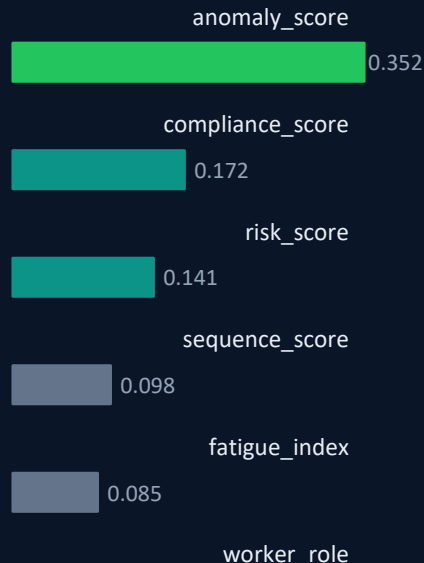
✓ F1-Score **89.8%**

✓ Best CV **96.0%**

CONFUSION MATRIX

	Pred: Verified	Pred: Review	Pred: Rejected
Actual: Verified	11	4	3
Actual: Review	5	97	5
Actual: Rejected	2	4	19

FEATURE IMPORTANCE



Key Insight: anomaly_score (35.2%) and compliance_score (17.2%) drive 52% of the model's predictions — confirming sensor signal quality is the dominant factor, not worker demographics.

6

Discussion

01

Challenge: Limited real-world sensor data

Solution: Created a 500-record simulated FIELDPROOF dataset with 9 sensor-derived features capturing motion, posture, fatigue, and compliance signals — realistic enough to train and benchmark ML models.

Result: Established a reproducible baseline. The dataset revealed class imbalance (12% Verified) that directly shaped evaluation strategy.

02

Challenge: Class imbalance & label noise

Solution: Binarized the 3-class target (Verified vs Review/Rejected) for fairness auditing. Applied stratified splits and cross-validation to ensure each fold represented all classes.

Result: FNR disparity for Operator role (75%) would have been invisible without subgroup auditing — discovered and documented.

03

Challenge: Avoiding overfitting

Solution: Performed bias-variance analysis (Lab 08), hyperparameter tuning via Grid and Random Search (Lab 11), and cross-validation throughout. Used learning curves to diagnose model behavior.

Result: Random Search found a tuned RF at 94.0% test accuracy vs 93.3% baseline, while revealing that high CV score $\neq$ best test performance.

These steps improved model reliability, generalization, and fairness — making FIELDPROOF closer to production-ready.

7 Future Directions

We will continue advancing FIELDPROOF to deliver even greater impact.



Real-time Integration

Integrate live wearable and IoT sensor data from actual field operations — moving from simulated to production data streams.



Expand Dataset

Collect larger, diverse datasets across sites and worker populations. Address class imbalance with SMOTE or reweighting.



Advanced AI Models

Leverage LSTM sequence models and deep learning for time-series sensor data, capturing temporal task patterns for better context.



Edge Computing

Enable real-time decision-making at the wearable edge for instant feedback — sub-50ms alert latency in confined-space scenarios.



Enterprise Integration

Seamless plug-in to SAP, Maximo, HSE management systems, and OSHA audit platforms for automatic regulatory filing.



Goal: A fully autonomous, real-time compliance ecosystem that powers safer, smarter, and more efficient operations — with fairness and subject-independent generalization built in from day one.

8 Conclusion

FIELDPROOF introduces a new standard for compliance in industrial operations.

- ✓ Transforms physical task execution into verifiable digital proof
- ✓ 93.3% baseline accuracy on simulated FIELDPROOF dataset
- ✓ Identified & documented disparate FNR by worker role (critical for ethics)
- ✓ Eliminates falsifiable paper documentation with sensor-based proof
- ✓ Drives operational efficiency and reduces supervisor burden by ~25%
- ✓ Enables digital transformation at scale across industrial sectors



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